

# Effects on Size Perception by Changing Dynamic Invisible Body Size

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## ABSTRACT

Only virtual hands and feet move synchronously with an observer's movement, inducing body ownership of an invisible body between them. However, it is unclear whether body ownership is also induced when the size of the invisible body is changed. In this study, we investigated whether body ownership is induced in large or small invisible bodies and whether size perception changes with the size of the invisible body. The results showed that body ownership was induced even if the size of the invisible body was changed, but the size perception did not change.

**Keywords:** Body ownership, invisible body, size perception

**Index Terms:** I.3.7 [Computer graphics]: Three-Dimensional Graphics and Realism—Virtual Reality; H.1.2 [Models and Principles]: User/Machine Systems—human factors

## 1 INTRODUCTION

A virtual body moves in synchronization with an observer's movement [1], or an experimenter touches a mannequin and an observer's body simultaneously with a stick [2], making the observer feel as if the body were their own body (illusory body ownership).

Body ownership can also be induced in bodies that differ in size from the innate body. van der Hoort et al. [3] showed that body ownership can be induced in small or large mannequins. Furthermore, they found that when the illusion is induced in the large body, objects appear smaller, while in the small body, objects appear larger. Surprisingly, changes in size perception also occur when a body is not presented at all. van der Hoort & Ehrsson [4] stimulated an empty space and the participant's body simultaneously with a stick and observed the participants watching the stick stroke in the empty space. This produced the illusion that the participant owned a large or small invisible body. As in the studies using large and small mannequins, the results showed that ownership of the large invisible body makes objects appear smaller, while ownership of the small invisible body makes objects appear larger.

The invisible body illusion by van der Hoort & Ehrsson [4] is static with visual-tactile stimuli, therefore it is not possible to move. In contrast, we developed the dynamic invisible body illusion in our previous study that allows active movement [5]. Only the hands and feet move synchronously with a participant's movement in a virtual environment, making an invisible space between the hands and feet feel as if it were their own body. On the other hand, it is

not clear whether our dynamic invisible body illusion also generates body ownership when the size of the invisible body is changed.

In this study, we investigate whether body ownership is induced in dynamic large or small invisible bodies. We also examine whether size perception changes depending on the size of the invisible body.

## 2 METHODS

**Participants.** 24 volunteers participated in the experiment. They had healthy vision and motor abilities. The sample size was calculated by G\*Power3.1 (repeated measures ANOVA, four conditions, medium effect size  $f = 0.25$ ,  $\alpha = 0.05$ , power  $(1 - \beta) = 0.8$ ). This study was conducted with the approval of the Ethical Committee on Human-subject Studies of Keio University.

**Stimuli and Conditions.** The participants wore motion capture suits, and their movements were tracked by 24 motion capture cameras (OptiTrack). Participants put on an HMD (HTC Vive Pro Eye) to observe a virtual environment created by Unity. An avatar with only gloves and socks was presented in the virtual environment (Figure 1), and the scale of the avatar was either large (255 cm tall) or small (113 cm tall). The small avatar's viewpoint was adjusted to the large avatar's viewpoint by placing a cube under the small avatar. Also, there were two conditions in which the gloves and socks moved synchronously with the participant's movement (S condition) and an asynchronous condition in which the gloves and socks moved with a 1s delay (A condition).



Figure 1: Large invisible body (left), Small invisible body (right)

**Learning Session.** The participants moved their bodies in the order of right hand, left hand, right foot, and left foot for 2 minutes according to the sound (1 kHz, 500 ms) presented through the headphones every 2 seconds. They were also instructed to look in a mirror when moving the gloves and socks.

**Size estimation task.** A 10, 20, or 30 cm cube was presented for 2 s at 1.5 m in front of the participant. The screen then went dark, and participants were instructed to adjust the distance between their hands to be the same size as the presented cube. Once the hand distance was determined, the participant called out to the experimenter, who pressed a key and recorded the distance between the hands. The width of the hands at this time was taken as the participant's estimated cube size. In one task, size estimates were made for all cube sizes (three trials in total). The order in which the

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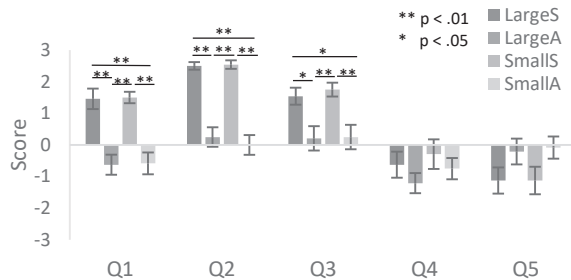
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cube sizes were presented was random. In addition, 12 different cube colors were available, and a cube of a different color was presented in all experimental trials.

**Procedures.** In the experiment, the participants practiced one block of the size estimation task, followed by one block of the pre-learning size estimation task (baseline). Then, after moving the gloves and socks for 2 minutes in the learning session, they again performed the size estimation task for one block, and finally, they completed the questionnaire. Participants answered five questions on a seven-level Likert scale ranging from -3 (strongly disagree) to +3 (strongly agree). Each question was presented in random order. A total of four trials of the procedure from the learning session to the questionnaire were conducted in random order (two avatar size conditions x two movement conditions x one repetition).

### 3 RESULTS

**Questionnaire.** Regardless of size, body ownership was stronger when the virtual gloves and socks moved in synchrony with the participant's movement than in the asynchronous condition (Q1 LargeS vs. LargeA:  $z = 3.58$ ,  $p < .001$ ,  $r = .73$ ; SmallS vs. SmallA:  $z = 3.56$ ,  $p < .001$ ,  $r = .73$ ; Figure 2). Similarly, the movement synchronizing the gloves and socks felt more like their own movement than the asynchronous movement (Q2 LargeS vs. LargeA:  $z = 4.14$ ,  $p < .001$ ,  $r = .85$ ; SmallS vs. SmallA:  $z = 4.04$ ,  $p < .001$ ,  $r = .83$ ). Also, when the movement was synchronized, participants felt as if a full body was in the space between the gloves and socks more than asynchronously (Q3 LargeS vs. LargeA:  $z = 2.65$ ,  $p = .036$ ,  $r = .80$ ; SmallS vs. SmallA:  $z = 3.16$ ,  $p < .001$ ,  $r = .64$ ). Q4 and Q5 were control questions, and there were no significant differences between conditions.



|    |                                                                             |
|----|-----------------------------------------------------------------------------|
| Q1 | It felt as if the space between the virtual gloves and socks was my body.   |
| Q2 | The movement of the virtual gloves and socks seemed to be my movement.      |
| Q3 | It seemed as if there was a full body between the virtual gloves and socks. |
| Q4 | I had two bodies at the same time.                                          |
| Q5 | It seemed as if the virtual gloves and socks had a will of its own.         |

Figure 2: Results of questionnaire

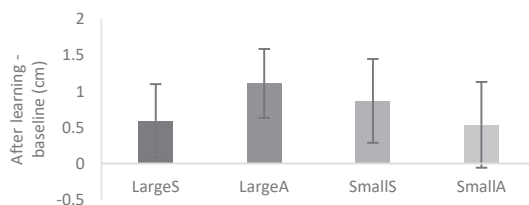


Figure 3: Results of size estimation task

**Size estimation task.** The change in size perception was analyzed by subtracting the results of the baseline trials from the results of each experimental trial. Results showed no significant differences between avatar sizes (LargeS vs. SmallS:  $z = -0.43$ ,  $p = 1.00$ ,  $r = .087$ ; Figure 3).

### 4 DISCUSSION

In this study, we investigated whether a larger or smaller invisible body generates body ownership. We also examined whether size perception changes with the size of the invisible body. The results showed that when the gloves and socks moved synchronously with the participant's movement, body ownership (Q1) was induced in the space between the gloves and socks in both the large and small body conditions. The results suggest that the invisible body illusion occurs regardless of the size of the body. The sense of agency (Q2) and the perception of the full body (Q3) also occurred regardless of the size of the body when the motion was synchronized. These results are similar to our previous invisible body study [5].

On the other hand, size perception did not change when the size of the invisible body changed. This is a different result from the static invisible body study [4]. In their study, there was a big difference between the size of the large invisible body (300-400 cm) and the small invisible body (60-80 cm). In this study, 170 cm was used as the standard, 1.5 times 170 cm as the large condition, and 170 divided by 1.5 as the small condition. The purpose was to control changes in the scale. However, it might have been better to make a big difference in avatar size, according to the static invisible body study [4].

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### REFERENCES

- [1] M. Gonzalez-Franco, D. Perez-Marcos, B. Spanlang, and M. Slater, "The contribution of real-time mirror reflections of motor actions on virtual body ownership in an immersive virtual environment," in *2010 IEEE Virtual Reality Conference (VR)*, 2010, pp. 111-114, doi: 10.1109/VR.2010.5444805.
- [2] V. I. Petkova and H. H. Ehrsson, "If I were you: Perceptual illusion of body swapping," *PLoS One*, vol. 3, no. 12, p. e3832, Dec. 2008, doi: 10.1371/journal.pone.0003832.
- [3] B. van der Hoort, A. Guterstam, and H. H. Ehrsson, "Being Barbie: The Size of One's Own Body Determines the Perceived Size of the World," *PLoS One*, vol. 6, no. 5, p. e20195, May 2011, doi: 10.1371/journal.pone.0020195.
- [4] B. van der Hoort and H. H. Ehrsson, "Illusions of having small or large invisible bodies influence visual perception of object size," *Sci. Rep.*, vol. 6, no. 1, p. 34530, Dec. 2016, doi: 10.1038/srep34530.
- [5] R. Kondo, M. Sugimoto, K. Minamizawa, T. Hoshi, M. Inami, and M. Kitazaki, "Illusory body ownership of an invisible body interpolated between virtual hands and feet via visual-motor synchronicity," *Sci. Rep.*, vol. 8, no. 1, pp. 1-8, Dec. 2018, doi: 10.1038/s41598-018-25951-2.